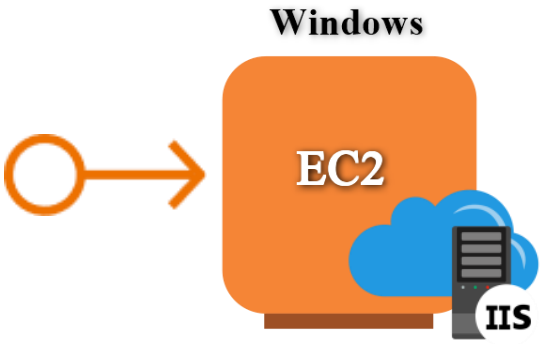
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

3.Configure Elastic IP Address to Windows Web Server in AWS



Santhu Mohammed M

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

Coimbatore.

Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

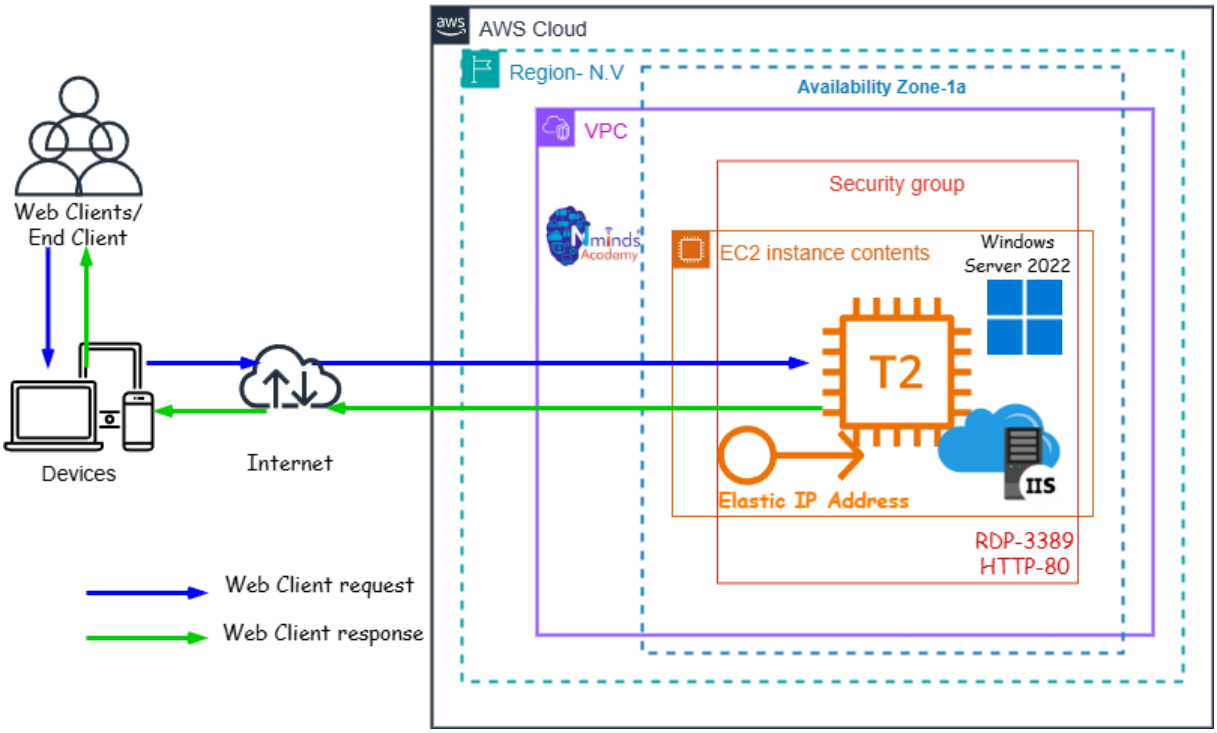
1. Highly Available Applications:
    - If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
    - If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
    - Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

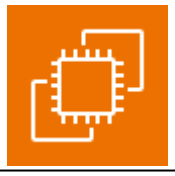
Topology

How to Configure the Elastic IP Address to Windows Web Server with AWS EC2



Used Services in AWS

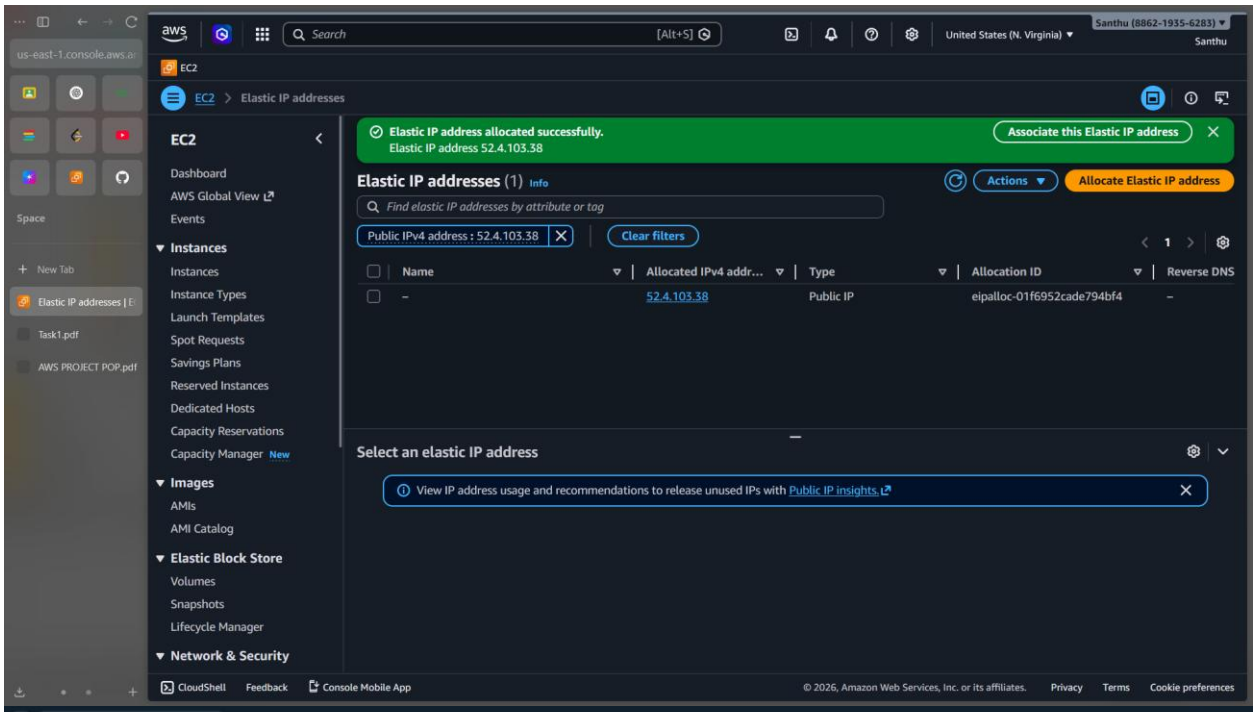
Used Services in Windows Server

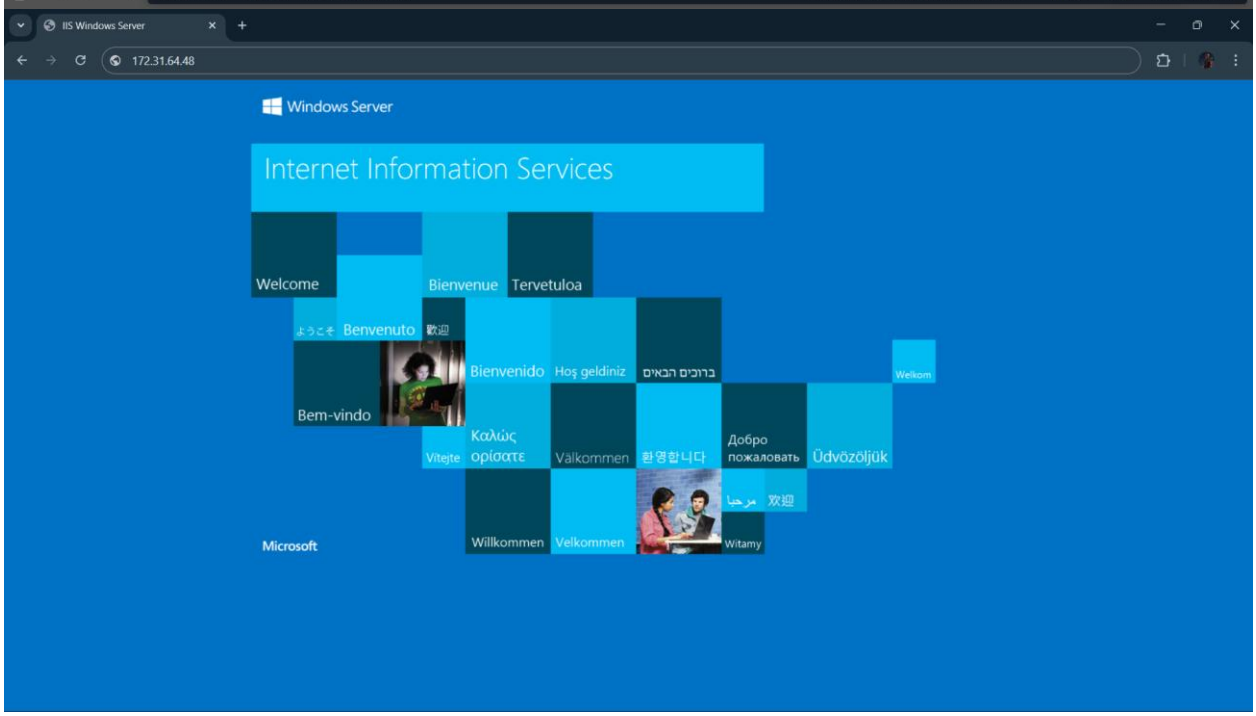


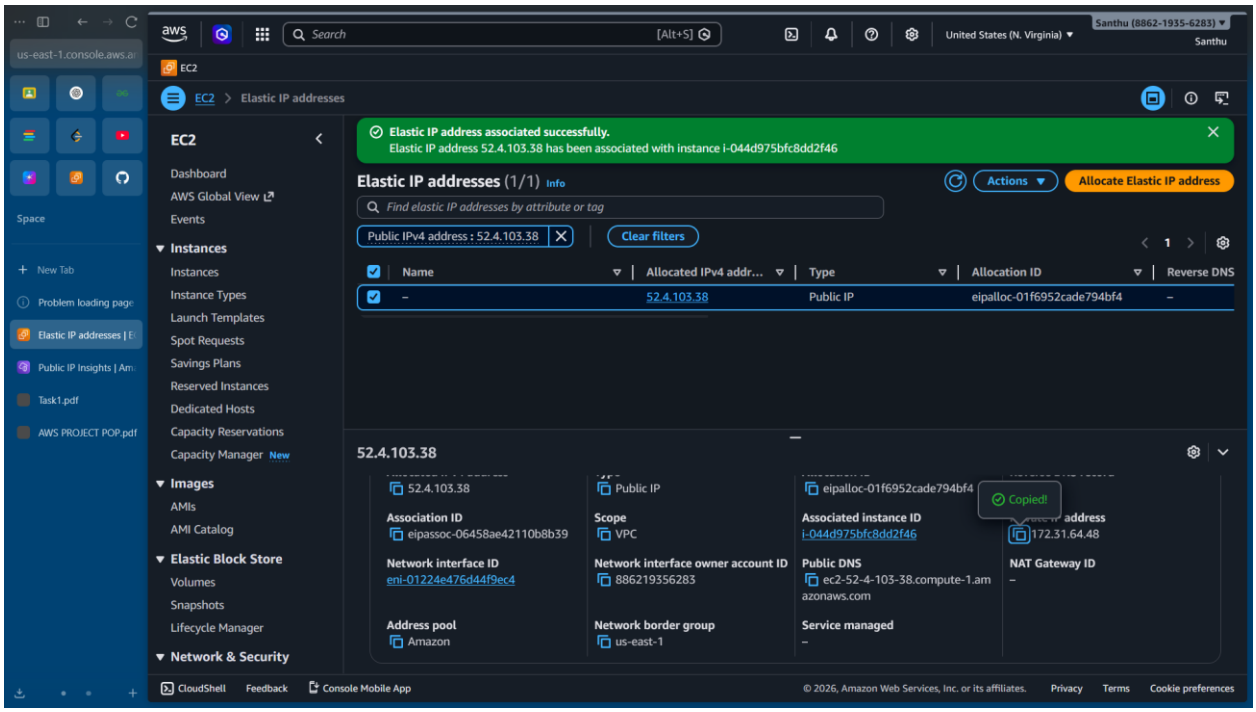
Amazon EC2

IIS- Internet Information Services

Amazon Elastic IP Address


Elastic IP address allocated successfully.
Elastic IP address 52.4.103.38
Associate this Elastic IP address
Elastic IP addresses (1) Info
Actions
Allocate Elastic IP address
Public IPv4 address : 52.4.103.38
Clear filters
Name
Allocated IPv4 addr...
Type
Allocation ID
Reverse DNS
52.4.103.38
Public IP
eipalloc-01f6952cade794bf4
Select an elastic IP address
View IP address usage and recommendations to release unused IPs with Public IP insights.


172.31.64.48
Windows Server
Internet Information Services
Welcome
Bienvenue
Tervetuloa
Benvenuto
Bienvenido
Hoş geldiniz
Bem-vindo
Welkom
Vítejte
Καλώς ορίσατε
Välkommen
Добро пожаловать
Üdvözöljük
Willkommen
Velkommen
Witamy
Microsoft


Elastic IP address associated successfully.
Elastic IP address 52.4.103.38 has been associated with instance i-044d975bfc8dd2f46
Elastic IP addresses (1/1) Info
Actions
Allocate Elastic IP address
Find elastic IP addresses by attribute or tag
Public IPv4 address : 52.4.103.38
Clear filters
Name
Allocated IPv4 addr...
Type
Allocation ID
Reverse DNS
52.4.103.38
Public IP
eipalloc-01f6952cade794bf4
52.4.103.38
Association ID
eipassoc-06458ae42110b8b39
Scope
VPC
Associated instance ID
i-044d975bfc8dd2f46
Copied!
172.31.64.48
Network interface ID
eni-01224e476d44f9ec4
Network interface owner account ID
886219356283
Public DNS
ec2-52-4-103-38.compute-1.amazonaws.com
NAT Gateway ID
Address pool
Amazon
Network border group
us-east-1
Service managed